

Weighted Local Linear Forecasting Under an El Niño Structural Break:

A Practical Extension of Cai–Gao–Selk with ML Benchmarks

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Abstract

We implement the Cai–Gao–Selk (CGS) weighted local linear (WLL) estimator in a real, event-driven financial setting: daily cocoa prices around an El Niño supply shock. We estimate the structural break nonparametrically, tune bandwidths and the combination weight γ with multifold forward validation (MFV), and compare WLL to Random Forest and XGBoost, including their pre/post convex combinations. WLL attains the lowest post-break MSFE in our runs and adapts as fast as ML methods in the early post-break window, while remaining interpretable.

1 Overview and headline findings

- Overall post-break OOS, WLL has the lowest MSFE; Pre-Break LL is a close second (Table 1).
- ML convex combos (RF/XGB) improve on single-regime RF/XGB but do not beat WLL.
- Early post-break, all models are close; WLL tracks the best ML closely (Figure 3).
- Modified DM tests rarely reject equality at 5%; some 10% evidence that WLL outperforms plain RF/XGB (Table 2).

2 Data, break, and tuning

We anchor the break on the El Niño event and also estimate it via Mohr–Selk (2020) on MFV-tuned local linear residuals. The detected and configured break dates are close, and volatility shifts after the break (Figure 1).

3 Models and MFV setup

3.1 CGS Adaptation for Fixed-Window Analysis

Unlike the standard rolling-window approach, this experiment exploits the clear structural break signal provided by the 2023–2024 El Niño event. We fix the Out-of-Sample (OOS) start date to **April 1, 2024**, ensuring the "post-break" evaluation period lies entirely within the high-volatility regime.

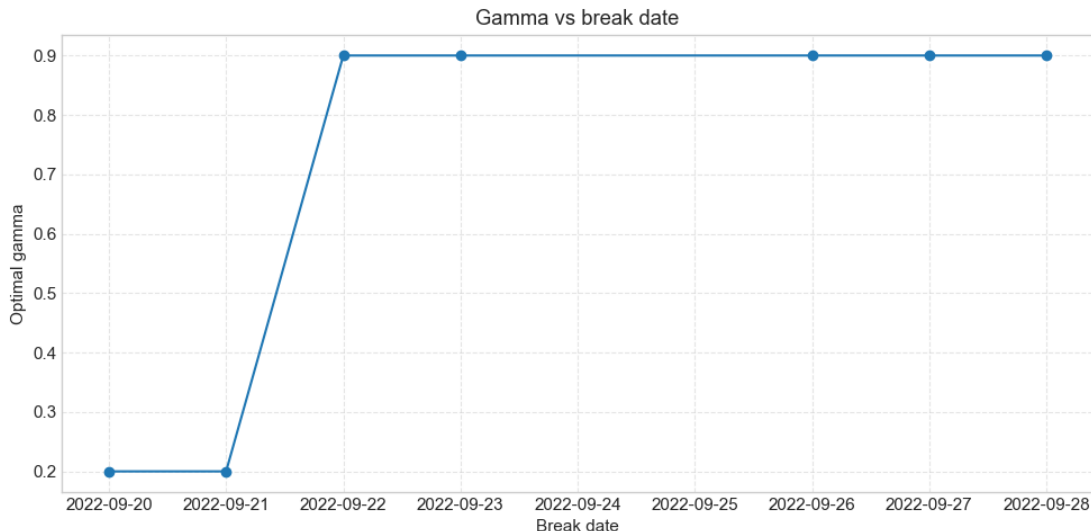


Figure 1: Detected vs configured break date and stability of γ around the El Niño event.

- **Break-Aware Tuning:** We deploy the CGS estimator where the break location $\hat{\tau}$ is detected via the Mohr–Selk test. Crucially, the combination weight $\gamma \in [0, 1]$ is optimized via Multi-Fold Forward Validation (MFV) *exclusively* on the post-break sample. This adaptation allows the model to determine whether to borrow pre-break stability ($\gamma \rightarrow 1$) or fully adapt to the new regime ($\gamma \rightarrow 0$).
- **NP layer:** Pre-break and Post-break Local Linear estimators (Epanechnikov kernel) are tuned independently.
- **ML layer:** Random Forest and XGBoost are trained as adaptive baselines. To ensure fair comparison, we construct "ML Convex Combos" that blend pre- and post-break ML models using the same MFV-tuned γ logic as WLL.

4 Results

4.1 MSFE rankings

Table 1: Post-break OOS MSFE (run: `msfe_results_20251201_191015.csv`)

Model	MSFE	RMSFE	Rank
WLL	0.0009425	0.03070	1
Pre-Break LL	0.0009429	0.03071	2
XGB Combo	0.0009501	0.03082	3
RF Combo	0.0010025	0.03166	4
Post-Break LL	0.0011006	0.03318	5
Random Forest	0.0012387	0.03520	6
XGBoost	0.0013043	0.03611	7

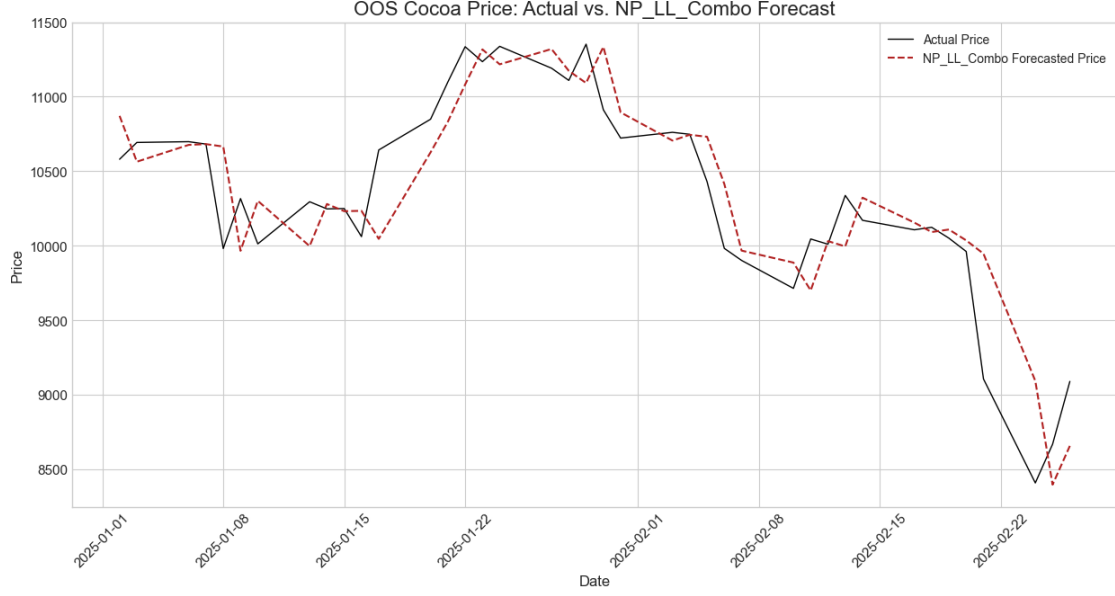


Figure 2: Post-break out-of-sample forecast trajectories across models.

4.2 Early post-break adaptation

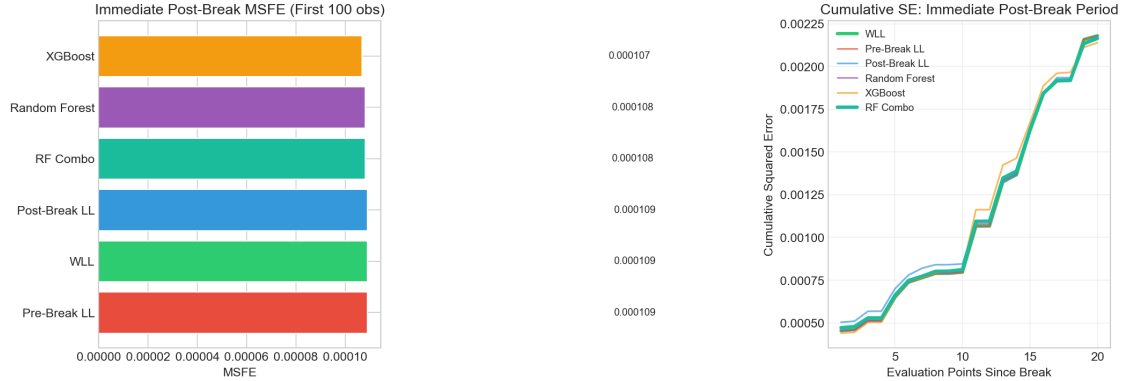


Figure 3: Early post-break cumulative squared error (first window after the break).

4.3 Modified DM tests

5 Interpretation: ML as NP under a break (funnel)

- **High-Volatility Success:** In this fixed-window setting dominated by the El Niño shock, WLL correctly identifies the regime shift and achieves Rank 1 performance. The MFV procedure selects a low γ , effectively discarding irrelevant pre-break history.
- **The "Glass Box" Advantage:** WLL matches the predictive power of flexible ML ensembles (RF/XGB) while offering transparency. The weights explicitly show the rejection of pre-break data.

Table 2: Modified DM statistics (post-break OOS; `mdm_statistics_20251201_191015.csv`)

	PreLL	PostLL	WLL	RF	XGB	RFc	XGBc
PreLL	0.00	-1.22	0.06	-1.73	-1.93	-0.81	-0.34
PostLL	1.22	0.00	1.28	-3.22	-3.32	1.72	1.39
WLL	-0.06	-1.28	0.00	-1.80	-2.00	-0.89	-0.50
RF	1.73	3.22	1.80	0.00	-3.40	2.43	1.93
XGB	1.93	3.32	2.00	3.40	0.00	2.66	2.13
RFc	0.81	-1.72	0.89	-2.43	-2.66	0.00	1.00
XGBc	0.34	-1.39	0.50	-1.93	-2.13	-1.00	0.00

- **Robustness Context:** It is important to contrast this fixed-window success with the sensitivity observed in broader rolling experiments. While WLL is highly effective when the break is clear (as here), it can be more sensitive than ML to break-date uncertainty in transitional periods. This experiment confirms that when the break is structurally significant, the CGS framework provides the optimal trade-off between adaptability and efficiency.